Chapter 30

Measurement Risk and Decision Rules

MEASUREMENT RISK

Measurements are frequently utilized in our daily activities. We make decisions on measurements that are made to accept/reject or buy/sell products and services. A company might budget research and development (R&D) costs based on specifications required or mandated by the customer or regulations. Medical professionals use measurement results to assess the outcome of diagnostic tests. Different industries utilize test method results by conducting defined or random sampling methods on individual parts, sub-assemblies, and assemblies to make conformance decisions. For example, a satellite to be deployed in Earth's orbit will undergo various environmental, structural, reliability, and other tests.

Imagine that a satellite that has been launched into space behaves in a way in which it is not designed to function. Suppose the satellite experiences a wobble, which causes connectivity problems at the receiving station on Earth. The cause of the wobbling is identified as use of a calibration provided with an incorrect measurement uncertainty, which thus resulted in an incorrect measurement conformance decision. For example, each of the load cells used to measure the fuel load stored in the satellite must include the confidence probability limit in its calibration. However, if a calibration provider does not have the measurement confidence due to incorrect assumptions, the load cell calibration may not be fit for the purpose. In this case, the result is a wobbling satellite requiring significantly more resources to fix the problem. A wobbly satellite is one example of many where measurement uncertainty, when not adequately accounted for, leads to significant engineering disasters, trillions of dollars in losses, and potentially the loss of human life. Fixing the faulty satellite is not as easy as hopping on a commercial flight and renting a car to get to it! These scenarios are played out daily in all fields of measurement, including the treatment of diseases and medical advances to save or prolong life.

Measurements made and reported for any application *must exhibit measurement confidence*, and the *decision made must be fit for the purpose*.

Measurement confidence comes from these "three pillars of measurement":

1. Measurement Uncertainty
2. Metrological Traceability
3. Measurement Decision Rules fit for the purpose, taking into account the acceptable risk

- Incorrect evaluation of measurement uncertainty (CMC) listed on their Scope of Accreditation (SoA)
- Incorrect application of decision rules to make conformance decisions
- Not understanding the conformance requirements of the customer, regulatory standards, or normative standards

Most importantly, making a conformance decision requires the engagement of the customer or end user. Traditionally, the consumer (customer) of calibration and test services relies on the expertise of the calibration and test provider, and the conformance decision relies on the manufacturer's specifications without taking the associated measurement uncertainty into account. Therefore, it becomes important to note the requirements of ISO/IEC 17025:2017: "When the customer requests a statement of conformity to a specification or standard for the test or calibration (e.g., pass/fail, in-tolerance/out-of-tolerance), the specification or standard and the decision rule shall be clearly defined. Unless inherent in the requested specification or standard, the decision rule selected shall be communicated to, and agreed with, the customer.

Note: For further guidance on statements of conformity, see ISO/IEC Guide 98-4."[22]

The customer must understand the capability of its test and calibration provider and its own conformance requirements before it can specify the test and calibration service to reduce their risk appetite. It must specify the decision rules required for the test and calibration provider that is making the conformance decision on a product. The test and calibration provider must understand the customer requirements for conformance decisions so that when the measurement result is reported with a confidence statement it considers:

- The proper evaluation measurement uncertainty
- Proof of metrological traceability
- Proper application of measurement decision rules to address the risk required in reporting any statements of conformity

TEST UNCERTAINTY RATIO (TUR)

Another decision rule that is quite frequently employed is the application of the *Test Uncertainty Ratio (TUR)*. This rule has its roots in the U.S. military specifications, and its application was listed in the MIL STD 45662 as early as the 1960s. The TUR has its limitations regarding conformance decisions and risk mitigation. In addition, there are different interpretations of how the TUR is calculated, which leads to further confusion.

Calculation of Test Uncertainty Ratio

Correctly calculating TUR is crucial because it is a commonly accepted practice when making a statement of conformity. It helps you analyze measurement risk and is often used in decision rules. Although TUR is defined in the ANSI/NCSL Z540.3 standard (NCSL International's *Handbook for the*

USL = Upper Specification Limit
Uncert. Dist = Uncertainty Distribution
LAL = Lower Acceptance Limit
UAL = Upper Acceptance Limit

A 4:1 TUR

In "Why a 4:1 TUR is not Enough," Zumbrun notes that several organizations and publications reference or insist on maintaining a 4:1 Test Uncertainty Ratio without understanding the level of measurement risk. The assumption is that if the lab performing the calibrations has standards at least four times better than what they are calibrating, everything is good. In Clause 5.3 b), ANSI/NCSL Z540.3, *Requirements for Calibration of Measuring and Test Equipment,* allows for use of a TUR equal to or greater than 4:1 when it is not practical to estimate the false accept risk of less than 2 %. That Clause then goes on to say that objective evidence of nonpracticability of this determination is expected, as in an agreement with the customer TUR use. The 4:1 TUR seems to be a fallback position that many industries have adopted, maybe because they did not understand or want to deal with guard bands. The assumption is that the higher the TUR, the higher the probability that the measuring equipment will have a probability of false accept (PFA) of less than 2 %, assuming that the measured reading is closer to the nominal value.[36]

Many fail to realize what is described in *Introduction to Statistics in Metrology*: Using a TUR assumes that all measurement biases have been removed from the measurement process and the measurements involved follow a normal distribution. If there are significant biases that cannot be removed, then the TUR will not account for the increased risk.[37]

CONCLUSION

Any decision rule used in making a conformance decision can affect the outcome of the application of the unit under test.

ISO/IEC 17025:2017 dictates that the decision rule must be agreed upon by the customer when a statement of conformity is to be made. The customer shall specify the tolerance requirements for conformity. If no conformance requirement is requested, then the requirement for decision rules is not necessary. Simply put, regarding decision rules, if you choose not to decide, you still have made a choice (shared risk). In this scenario, the consumer can bear as much as 50 % of the measurement risk. If the UUT is used to propagate measurement uncertainty further, Type II error is more likely to occur. However, the test and calibration providers are still obligated to report the measurement result and its associated measurement uncertainty.

Part of any risk mitigation strategy should consider the following:

- Selecting a calibration supplier that offers the smallest measurement uncertainty

- Utilizing the appropriate reference standards
- Improving reliability by managing calibration intervals
- Monitoring of standards using control charts
- Continual improvement of the calibration processes
- Using the appropriate decision rules

This chapter has provided guidance on applying and interpreting decision rules to meet customer, regulatory, and normative requirements for reporting statements of conformity and the associated measurement risk. Annex A gives an example of application of both methods 5 and 6 guard bands.

ANNEX A

	Nominal Value	1000
	Lower Specification Limit	998
	Upper Specification Limit	1002
	Measured Value	1001.5
	Std. Uncert. (k=1)	0.23
Z540.3 Method 5 (GB LSL & GB USL)	Test Uncertainty Ratio (TUR)	4.44
	Upper Guardband	1001.5376
	Lower Guardband	998.4624
	Total Guardband Risk	1.319%
	Upper Guardband Risk	1.319%
	Lower Guardband Risk	0.000%
Z540.3 Method 6 (LM6 & UM6)	Test Uncertainty Ratio (TUR)	4.44
	0.38 × ln(TUR) – 0.54	0.026544558
	$M_{2\%}$	1.310%
	Acceptance Limit	1.994100589
	$U_{2\%}$ Acceptance Limit	1001.994101
	$L_{2\%}$ Acceptance Limit	998.0058994
	$U_{2\%}$ PASS OR FAIL	PASS
	$L_{2\%}$ PASS OR FAIL	PASS
	TUR For Method 6	4.44

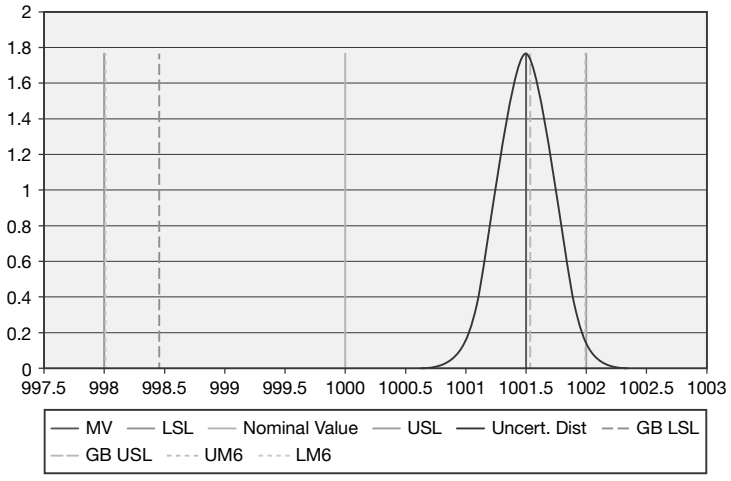


Figure 30.28 *ANSI/NCSL Z540.3 Handbook* Methods 5 and 6 guard bands.

Source: Reproduced by permission from Morehouse and E = mc³ Solutions.[38]

Figure 30.28 shows application of both methods 5 and 6 guard bands per NCSL International's ANSI/Z540.3 *Handbook*. The example uses a 1,000 lbf load cell with a resolution of 0.1 lbf and a 0.2 % of full-scale tolerance requirement. The standard uncertainty, which includes the Calibration and Measurement Capability uncertainty parameter and contributions from the unit under test from resolution, repeatability, environmental, and other error sources, is 0.23 lbf.

When calculated correctly, the TUR is 4.44:1. The methods shown in Figure 30.28 take into account the location of the measurement of 1001.5. If either of these methods were to be used for a conformity assessment, the instrument would pass, assuming the requirement was a PFA of less than 2 %; however, the acceptance limits are different. Method 5 is much more conservative, meaning a higher probability of false reject (PFR) and less risk for the end-user. If the value observed for the unit under test was read anywhere from 1001.6–1001.9, it would fail using method 5 and pass using method 6.